



AYE
TECH HUB

• PROTASTRUCTURE • LESSON 03

Structural Design Principles

Loads · Combinations · Tributaries · Frame Behaviour

Lesson 03 of 30 | ProtaStructure Program

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- The 5 load types every structural engineer must know cold
- Eurocode and ACI load combinations — the numbers explained
- Tributary area: how floor loads reach the columns below
- Sway vs non-sway frames and why it matters for column design
- How ProtaStructure models all these concepts automatically

Structural Loads — What Acts on Every Building

Before you model a single beam or column in ProtaStructure, you must understand what forces act on buildings. Structural engineering is **the science of safely transferring loads from the top of a building down to the ground**. Every element you design must carry its share of these loads without failing or deforming excessively.

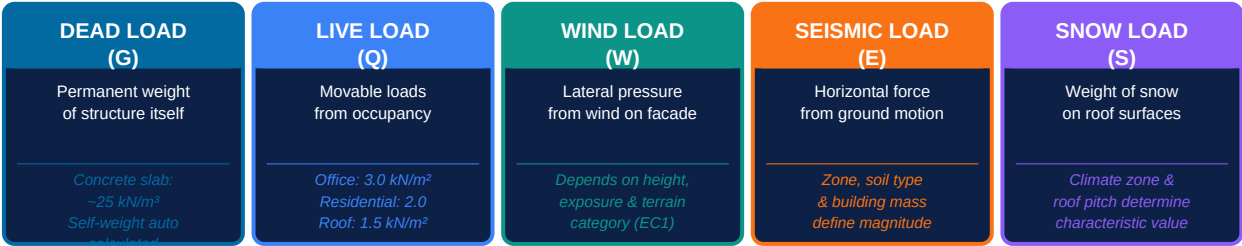


Figure 1 — The 5 Structural Load Types

Load Type	Symbol	Typical Values	In ProtaStructure
Dead Load (Permanent)	G or DL	RC slab 150mm: ~3.75 kN/m² Finishes (tiles+screed): ~2.0 kN/m² Partitions: ~1.0–2.0 kN/m²	Auto-calculated from element dimensions and unit weight. Manually add superimposed dead load per floor.
Live Load (Variable)	Q or LL	Residential: 2.0 kN/m² Office: 3.0 kN/m² Assembly (stadium): 5.0 kN/m² Storage: 7.5+ kN/m²	Input as distributed load per floor slab. Set as "Live" load type in load definition.
Wind Load	W or Wk	Generated using ASCE 7 or EC1 wind wizard. Typical 0.5–2.0 kN/m² on facade.	Use the Wind Load Generator (EN 1991-1-4 or ASCE 7) under Loads menu.
Seismic Load	E or Ek	Depends on seismic zone, soil, building weight and period.	Use the Seismic Load Generator — input zone, soil class and importance factor.

Self-weight is automatic: ProtaStructure calculates the self-weight of all structural elements (beams, columns, slabs, walls) automatically from their dimensions and the material density you assign (RC = 25 kN/m³). You do not need to input these manually — only superimposed dead loads (finishes, partitions, services) must be added by the engineer.

Tributary Area and Load Combinations

Before the computer calculates anything, you need to understand **how distributed floor loads are collected and transferred to columns**. This is the concept of **tributary area** — the floor area whose load is attributed to each structural support.

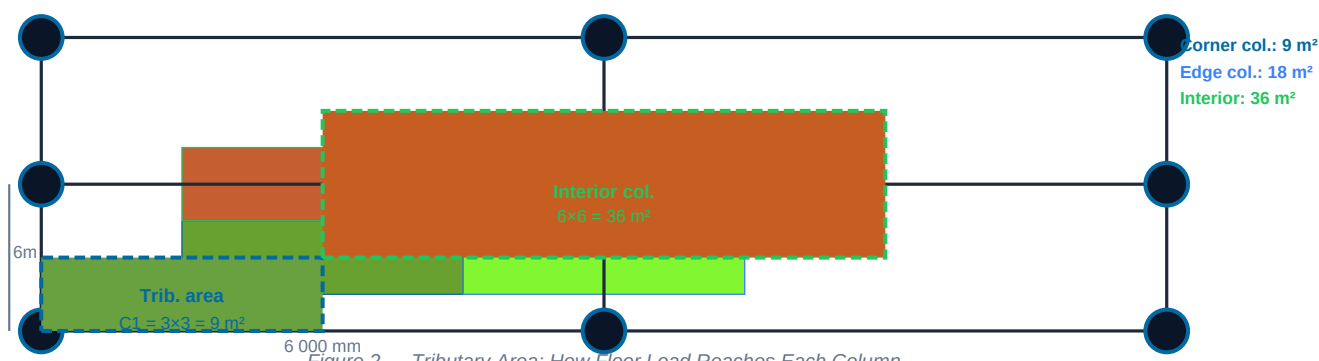


Figure 2 Tributary Area: How Floor Load Reaches Each Column

Tributary area rule: Each column carries the floor load from the half-span in every direction. A corner column carries $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ of one bay. An edge column carries $\frac{1}{2}$ of one bay. An interior column carries one full bay (all four half-spans). This is why interior columns carry 4× the load of corner columns in a regular grid — they must be larger in section.

Load Combinations — Eurocode 2 and ACI 318

Combination	Eurocode 2 Expression	ACI 318 Expression	Purpose
Gravity (max)	1.35G + 1.5Q	1.2D + 1.6L	Primary design case — maximum gravity load
Gravity + Wind	1.35G + 1.05Q + 1.5W	1.2D + 1.0L + 1.0W	Combined gravity and wind — governs lateral design
Gravity + Seismic	1.0G + 0.3Q + 1.0E	1.2D + 1.0L + 1.0E	Seismic design — governs in earthquake zones
Serviceability (SLS)	1.0G + 1.0Q (characteristic)	1.0D + 1.0L	Deflection and crack width checks — no safety factors
Uplift (wind)	0.9G + 1.5W (upward)	0.9D + 1.0W	Checks that wind does not lift the roof or overturn building

Factor	EC2 Symbol	Why >1.0?	Example
Dead load factor	$\gamma_G = 1.35$	Accounts for variability in material density and construction dimensions	Concrete density may be higher than assumed; slab may be thicker
Live load factor	$\gamma_Q = 1.5$	Accounts for the unpredictability of occupancy — loads can exceed characteristic	Office floor designed for 3.0 kN/m² may occasionally carry 4.0 kN/m²
Combination factor ψ	$\psi = 0.7$ (office Q)	Reduces live load when combined with wind — full live + full wind simultaneously is unlikely	Maximum live AND maximum wind at same time is a low-probability event

Load Path and Frame Behaviour

Understanding the **load path** — how gravity forces travel from where they are applied (roof, floors) down to where they are resisted (foundation, soil) — is the single most important concept in structural engineering. Everything else follows from this.

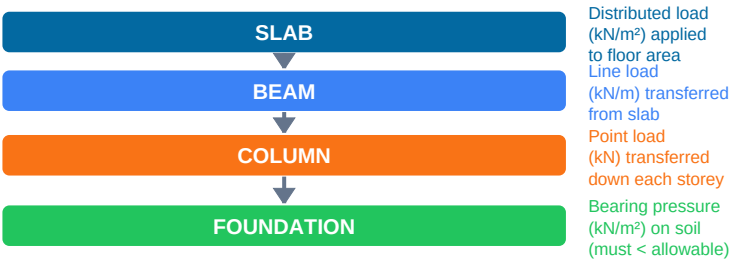


Figure 3 — Structural Load Path: From Slab to Foundation

Sway vs Non-Sway Frames

Frame Type	Definition	When It Occurs	Design Implication
Non-Sway Frame	Lateral stiffness is high enough that horizontal displacement under load is negligible ($< H/500$)	Stocky columns, many shear walls, braced frames, low-rise buildings	Simplified column design — second-order (P-Delta) effects can be ignored
Sway Frame	Significant lateral displacement occurs under horizontal loads — columns lean with the load	Slender columns, no shear walls, unbraced frames, tall buildings	Full second-order analysis required — P-Delta moments increase column bending significantly

How ProtaStructure Applies These Principles

Concept	ProtaStructure Feature	Where to Find It
Dead load (self-weight)	Automatically computed from element dimensions $\times$ material density	Analysis $\rightarrow$ Run Analysis — shown in results as "SW" load case
Superimposed dead + live loads	Input as area loads directly on slab elements after defining in Load Definition	Input menu $\rightarrow$ Loads $\rightarrow$ Slab Loads — assign kN/m^2 per storey
Wind load generation	Automated wind load wizard using EN 1991-1-4 or ASCE 7	Input menu $\rightarrow$ Loads $\rightarrow$ Wind Loads $\rightarrow$ Run Wizard
Seismic load generation	Seismic wizard: input zone, soil class, S_s , S_1 — generates equivalent static or dynamic loads	Input menu $\rightarrow$ Loads $\rightarrow$ Seismic Loads $\rightarrow$ Run Wizard
Load combinations	Pre-set EC2/ACI combinations or user-defined — applied automatically in analysis	Define $\rightarrow$ Load Cases & Combinations — review and modify if needed
Tributary area	ProtaStructure FEM solver distributes slab loads to supporting beams and columns automatically	Run Analysis $\rightarrow$ post-processing shows column axial forces from tributary area
Sway check	Checks sway sensitivity (α_{cr}) per EC2 Cl. 5.8 — classifies as sway or non-sway	Design results $\rightarrow$ Column Design $\rightarrow$ Sway Classification in calculation output

What comes next — PROTA-004: Grid and Level Setup — creating the complete structural grid and storey levels in ProtaStructure for a real building project, setting material grades, and configuring the analysis model before modelling any elements.